GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
Team Members: Annalakshmi R, Ashvitha I, Dharshini P, Joshna Amirthaa Mariappan

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WEEK 8 – CORRELATION ANALYSIS AND DIMENSIONALITY REDUCTION

During the eighth week, our team explored the relationships between different numerical variables and investigated dimensionality reduction techniques.

We began by performing correlation analysis to understand how variables were related to one another. We examined the correlation between departure delay and arrival delay and explored other numerical relationships within the dataset.

Using statistical calculations and visualization techniques, we created correlation summaries and examined the strength and direction of relationships between variables. We discussed how correlation values range from −1 to +1 and how they indicate positive, negative, or weak linear relationships.

The team then explored Principal Component Analysis (PCA), an unsupervised dimensionality reduction technique used to reduce the number of variables while retaining as much information as possible.

We studied the concepts of principal components, explained variance, and the importance of standardizing variables before applying PCA. We also discussed how the first principal component captures the maximum possible variance, followed by subsequent components.

We examined the usefulness of PCA in handling high-dimensional datasets and simplifying complex information for visualization and analysis.

By the end of the week, we had developed an understanding of correlation analysis and PCA and identified how these techniques could support the interpretation of flight delay patterns.